Stat 450: Case Studies in Statistics
Linear Mixed Effects (Random Effects) Models

Today you will learn to

- recognize when a linear mixed effects model is appropriate (compared to a linear fixed effects model)
- identify the different parts of the model (model matrices, variances, etc)
- implement linear mixed effects models in R (writing formulae, extracting model matrices, looking at summary, etc)
- use a linear mixed effects model to test a hypothesis

Kaggle Data: Do Bounce Times Depend on Age?

https://www.kaggle.com/ojwatson/mixed-models

Important website measure:
A bounce rate is a measure of how quickly someone leaves a website.

Experiment involving a recipe website

- 3 locations randomly chosen in each of 8 randomly chosen counties in England, 20 subjects randomly chosen in each.
- $3 \times 8 \times 20 = 480$ subjects
- Subjects used search engine to query for something they want to eat that evening. The search engine listed target website first.
- Subjects who clicked on the target website had bounce times recorded.

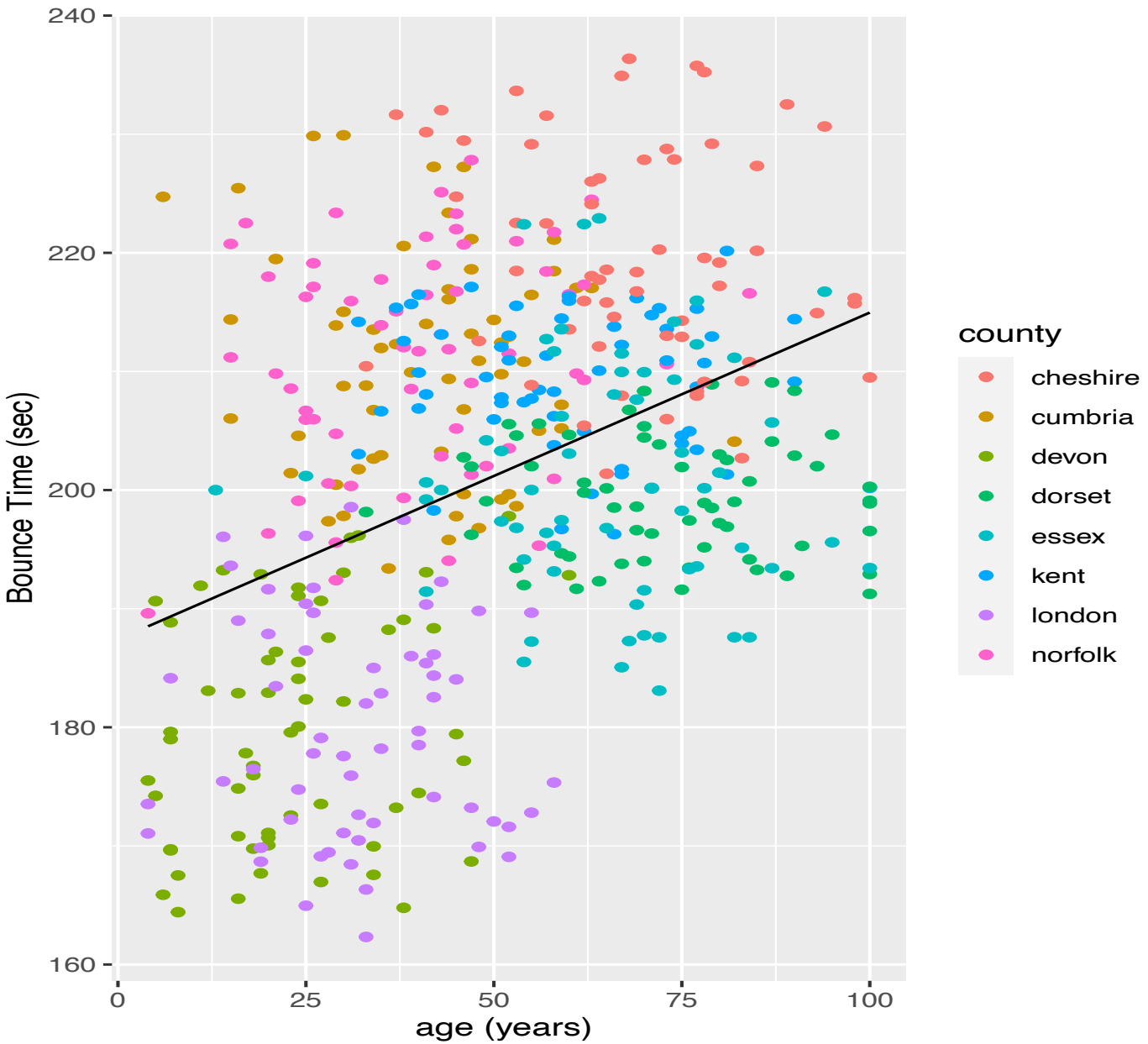
Question: are younger individuals more likely to leave the website sooner?

Bounce Time Data

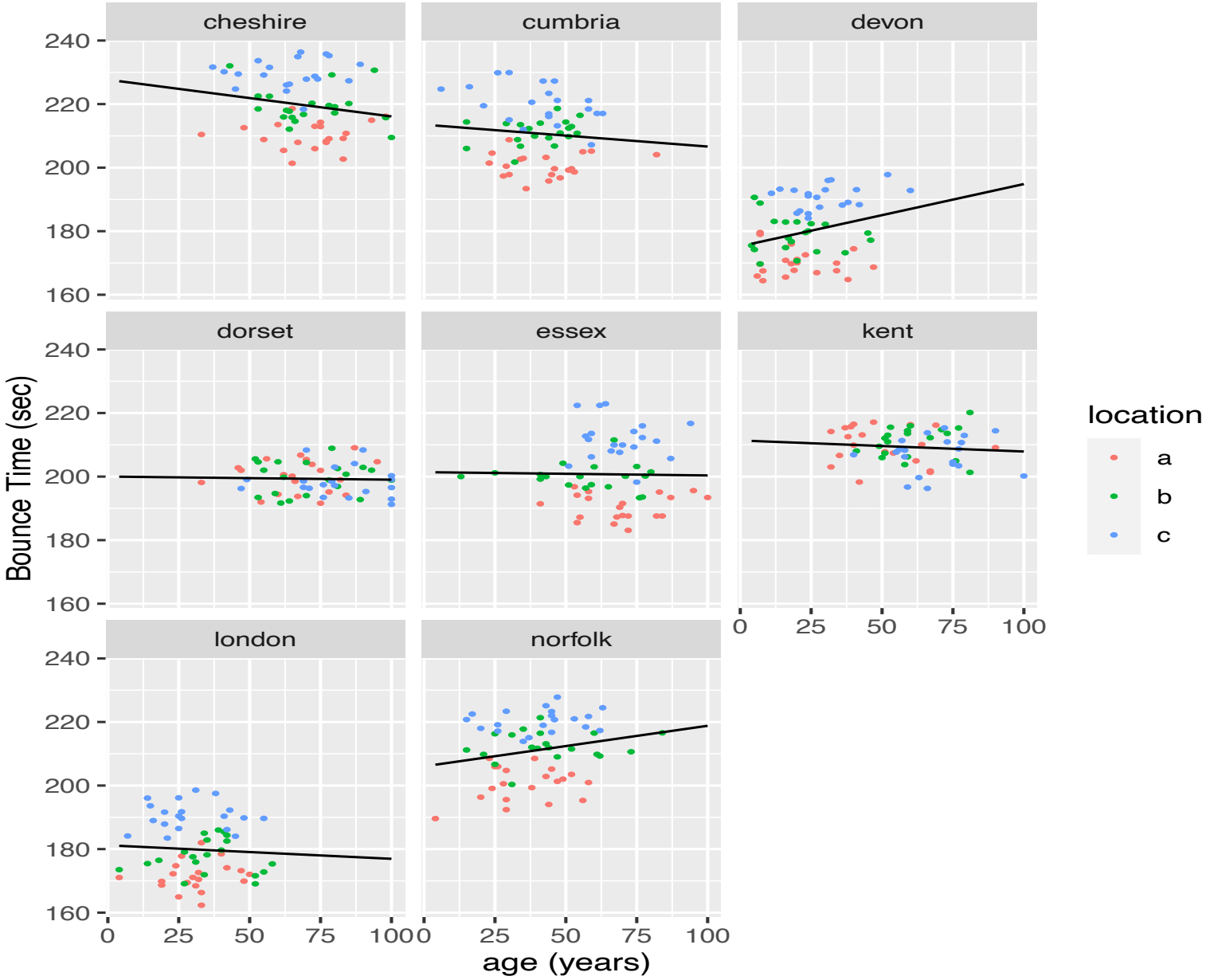
	bounce_time <dbl>	age <int>	county <chr>	location <chr>
1	165.5485	16	devon	a
2	167.5593	34	devon	a
3	165.8830	6	devon	a
4	167.6855	19	devon	a
5	169.9597	34	devon	a
6	168.6887	47	devon	a

8 counties, 3 locations in each county, 20 subjects from each location

Bounce Time versus Age with least squares line DON'T DO THIS!



Better: Bounce Time versus Age: 8 counties, 3 locations (a,b,c) per county



Linear model(s) for the data - omitting location (!)

Let

- $B_{c,i}$ = the bounce time of person i in county c
- $x_{c,i}$ = the age of person i in county c

$$B_{c,i} = \alpha_c + \beta_c \times x_{c,i} + \epsilon_{c,i}$$

In class, we know we can:

- fit 8 counties separately via regression (lm), with $\text{var}(\epsilon_{c,i}) = \sigma_c^2$
- fit one big lm for all 8 counties (separate intercepts and slopes) with $\text{var}(\epsilon_{c,i}) = \sigma^2$

But here: we should use linear mixed effects (random effects) model here.

Random effects model

Why?

- One slope and intercept won't work for all of England.
- There is correlation within county, even within location, so points in scatterplot aren't independent across counties.
- 8 counties were randomly chosen of England's 27 counties and the research question is for England, not the 8 counties.

Treat the "true" intercepts and slopes from each county as random variables.

$$B_{c,i} = \alpha_c + \beta_c \times x_{c,i} + \epsilon_{c,i}$$

But: sometimes also used if we'd looked at all 28 counties, to "borrow info" across counties.

Random Effects

$$B_{c,i} = \alpha_c + \beta_c\, x_{c,i} + \epsilon_{c,i} \quad \epsilon_{c,i} \sim \mathrm{N}(0, \sigma^2)$$

$$(\alpha_c, \beta_c)^T \text{ are random iid vectors}$$

independent of the $\epsilon_{c,i}$s

$$\mathrm{E}(\alpha_c) = a, \quad \mathrm{var}(\alpha_c) = \sigma_\alpha^2, \qquad \mathrm{E}(\beta_c) = b, \quad \mathrm{var}(\beta_c) = \sigma_\beta^2$$

$$\mathrm{cov}(\alpha_c, \beta_c) = \sigma_{\alpha,\beta}$$

Usually assume normality.

Recall question: are younger individuals more likely to leave the website sooner?

Test: $H_0: \quad b = 0 \quad$ versus $\quad H_a: \quad b > 0.$

Re-write model in terms of fixed and random effects

What we have:

$$B_{c,i} = \alpha_c \; + \; \beta_c \, x_{c,i} \; + \; \epsilon_{c,i}$$

$$\mathrm{E}(\alpha_c) = a, \quad \mathrm{var}(\alpha_c) = \sigma_\alpha^2, \qquad \mathrm{E}(\beta_c) = b, \quad \mathrm{var}(\beta_c) = \sigma_\beta^2$$

$$\mathrm{cov}(\alpha_c, \beta_c) = \sigma_{\alpha,\beta}$$

Rewrite:

$$B_{c,i} = a \; + \; \alpha_c^* \; + \; (b + \beta_c^*) \, x_{c,i} \; + \; \epsilon_{c,i}, \quad \epsilon_{c,i} \sim \mathrm{N}(0, \sigma^2)$$

with

$$\mathrm{E}(\alpha_c^*) = 0, \quad \mathrm{var}(\alpha_c^*) = \sigma_\alpha^2, \qquad \mathrm{E}(\beta_c^*) = 0, \quad \mathrm{var}(\beta_c^*) = \sigma_\beta^2$$

$$\mathrm{cov}(\alpha_c^*, \beta_c^*) = \sigma_{\alpha,\beta}$$

Model Matrices

$$\begin{pmatrix} B_{1,1} \\ B_{1,2} \\ \vdots \\ B_{2,1} \\ B_{2,2} \\ \vdots \end{pmatrix} = \begin{pmatrix} 1 & x_{1,1} \\ 1 & x_{1,2} \\ \vdots & \vdots \\ 1 & x_{2,1} \\ 1 & x_{2,2} \\ \vdots & \vdots \\ & \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} +$$

$$\begin{pmatrix} 1 & x_{1,1} & 0 & 0 & \cdots \\ 1 & x_{1,2} & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots \\ 0 & 0 & 1 & x_{2,1} & \cdots \\ 0 & 0 & 1 & x_{2,2} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \cdots \end{pmatrix} \begin{pmatrix} \alpha_1^* \\ \beta_1^* \\ \\ \alpha_2^* \\ \beta_2^* \\ \\ \vdots \end{pmatrix} + \epsilon$$

Two Model Matrices! **X** and **Z**!

$$\mathbf{Y} = \mathbf{X} \times \begin{pmatrix} \text{vector} \\ \text{of} \\ \text{expected} \\ \text{values} \end{pmatrix} + \mathbf{Z} \times \begin{pmatrix} \text{vector} \\ \text{of} \\ \text{zero mean} \\ \text{random} \\ \text{variables} \end{pmatrix} + \epsilon$$

$$= \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \epsilon$$

X and β: fixed effects
Z and **u**: random effects

Exercise: consider: what are **X**, **Z**, β and **u** for

1. random intercept, no slope (slope = 0)
2. random intercept, non-random slope - same for all counties?
3. random slope, non-random intercept - same for all counties?

Standard mixed effects model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\epsilon}$$

$$\mathbf{Y} \in \Re^n, \quad \boldsymbol{\beta} \in \Re^L, \quad \mathbf{u} \in \Re^J, \quad \boldsymbol{\epsilon} \in \Re^n$$

So $\mathbf{X}$ is $n \times L$ and $\mathbf{Z}$ is $n \times J$.

$$\mathrm{E}(\mathbf{u}) = 0, \quad \mathrm{E}(\boldsymbol{\epsilon}) = 0, \quad \mathrm{cov}\begin{pmatrix} \mathbf{u} \\ \boldsymbol{\epsilon} \end{pmatrix} = \begin{bmatrix} \Sigma_u & 0 \\ 0 & \Sigma_\epsilon \end{bmatrix}.$$

- the covariance matrix of $\mathbf{u} = \Sigma_u$
- the covariance matrix of $\boldsymbol{\epsilon} = \Sigma_\epsilon$.
- $\mathbf{u}$ and ϵ are uncorrelated
- typically, $\Sigma_\epsilon = \mathrm{diag}(\sigma^2)$.

We observe data $\mathbf{Y}$ and know $\mathbf{X}$ and $\mathbf{Z}$. We don't observe $\mathbf{u}$, ϵ. We don't know $\boldsymbol{\beta}$.